

Revised Load Interconnection System Impact Study Report

For Queue Project Q22-2131

Plantation 138 kV
June 24, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2131. The Interconnection Customer proposes a 166.5 MW large load (data center) facility interconnecting at the Plantation 138 kV substation in the Pennsylvania area. The request was received on July 10, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 476 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$8,382,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
INDUSTRIAL-CHIQUITA 138 kV	IRVINE-CHIQUITA 345 kV	121.5	194	8.8
INDUSTRIAL-PEQUOT LAKES SUBSTATION 138 kV line	CHIQUITA SUBSTATION-IRVINE 345 kV	111.1	185	8.8
KEENEY-DELL CRK 138 kV Line	SEMINOLE-MILLTOWN 138 kV Line	108.8	288	6.9
PLANTATION-SILVER SIDE RD 138 kV L	ORANGE PARK-NARROWS 138 kV L	106.1	288	51.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CHIQUITA-INDUSTRIAL 138 kV Line	Rebuild line with higher-capacity conductor	\$3,772,000	47% (\$1,780,000)
INDUSTRIAL-PEQUOT LAKES SUBSTATION 138 kV line	Rebuild line with higher-capacity conductor	\$4,854,000	47% (\$2,291,000)
KEENEY-DELL CRK 138 kV Line	Reconductor line	\$567,000	12% (\$65,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SILVER SIDE RD-PLANTATION 138 kV Line	Reconductor line	\$5,240,000	81% (\$4,246,000)
	Total allocated to project		\$8,382,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.